including tubercule bacilli and spores. It is an effective sterilizing agent because it rapidly penetrates packing materials, even plastic wrappers. To be potent, however, the humidity and temperature must be carefully controlled within narrow limits.

It is highly toxic on contact with the skin or mucous membrane. Materials that have been sterilized with ethylene oxide must be set aside in detoxification chambers for a few days to allow the gases to dissipate. It is frequently used to sterilize heart lung machines and plastic items like catheters.



Recently vapour – phase hydrogen peroxide has been used to decontaminate biological safety cabinets.

Formaldehyde

It is highly bactericidal. Formaldehyde is used as 40% formalin with humidity at around 50%. It causes irritation. It is used occasionally to fumigate rooms and disinfect respirators.

Betapropiolactone (BPL)

This is occasionally employed as a sterilizing gas in the liquid form. It has been used to sterilize vaccines, tissue grafts, surgical instrument and enzyme as a sterilants of blood plasma, water, milk and as a vapour – phase disinfectant in enclosed spaces, short-term inhalation exposure to betapropiolactone causes server irritation of the eyes, nose, throat and respiratory tract in humans.

BPL decomposes to an inactive form after several hours and is therefore not difficult to eliminate. It destroys microorganisms more readily than ethylene oxide but does not penetrate materials well and may be carcinogenic. For these reasons, BPL has not been used as extensively as EtO.

3.5 Evaluation of Antimicrobial Chemical Agents

Testing of antimicrobial agents is a complex process regulated by two different federal agencies.

The U.S. Environmental Protection Agency regulates disinfectants, where as agents used on humans and animals are under the control of the Food and Drug Administration.

Testing of antimicrobial agents often begins with an initial screening test to see if they are effective and at what required concentrations.

Laboratory techniques for the evaluation of antimicrobial chemical agents are conducted by one of the following three general procedures. In each procedure, the chemical agent is tested against a specified microorganisms referred to as the test organism.

Chemical Methods of Microbial Control

i. Chemical agents are used on living tissue (as antiseptics) and on inanimate objects (as disinfectants).

ii. Few chemical agent achieve sterility.

Evaluating a disinfectant

1. The phenol coefficient is the comparison of one chemical's disinfecting action with that of phenol, applied for the same length of time on the same organisms under identical conditions.

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